

COMPANION
to

ChatGPT Teaches TikZ

Chapter 3: Connecting Nodes (B)

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3.1 What This Companion Adds

Chapter 3 introduced four families of connections: straight, curved, labeled, and right-angled. This Companion develops the judgment needed to choose among them.

The central rule is simple: use the least complicated connection that shows the intended relationship clearly. A straight line is normally best. Add a bend, a label, or a right angle only when it communicates something useful or prevents confusion.

Design

Connections express structure. Their direction, route, and label should help the reader understand the diagram; they should not be chosen merely because they look elaborate.

3.2 A Connection Vocabulary

Form	Use
(A) -- (B)	A direct straight connection.
(A) to (B)	A flexible connection that can receive bend options.
-{Stealth}	A modern arrow tip showing direction.
bend left	Curves a connection to the left.
bend right	Curves a connection to the right.
node[midway]	Inserts a label halfway along a path.
node[near start]	Places a label near the first endpoint.
node[near end]	Places a label near the second endpoint.
-	Moves horizontally and then vertically.
-	Moves vertically and then horizontally.

The examples use the modern arrow library, loaded in the preamble with

```
\usetikzlibrary{arrows.meta}
```

3.3 Straight Connections

Use `--` when two nodes can be connected directly.

```
\begin{tikzpicture}
  \node[draw] (A) at (0,0) {$A$};
  \node[draw] (B) at (3,0) {$B$};
  \draw (A) -- (B);
\end{tikzpicture}
```

See



TikZ normally ends the line at the node boundaries. The source names the objects, and TikZ determines the exact boundary points.

3.3.1 Adding Direction

```
\draw[-{Stealth}] (A) -- (B);
```

Reversing the endpoint names reverses the direction:

```
\draw[-{Stealth}] (B) -- (A);
```

3.4 Curved Connections

Curves are useful when two relationships join the same pair of objects or when a straight edge would cross important material.

```
\begin{tikzpicture}
  \node[draw] (A) at (0,0) {$A$};
  \node[draw] (B) at (4,0) {$B$};

  \draw[-{Stealth}] (A)
    to[bend left=22] (B);

  \draw[-{Stealth}] (B)
```

```
to[bend left=22] (A);
\end{tikzpicture}
```

See



Each path bends to its own left, so the two arrows separate naturally.

3.4.1 Choosing the Bend

A value such as `bend left=15` creates a gentle curve; `bend left=40` creates a much stronger one. Increase the bend only as far as needed to separate the paths.

Common Mistake

Do not replace every straight edge by a curve. Unnecessary curves make the reader search for a reason that may not exist.

3.5 Labels on Connections

A label on a path is itself a node.

```
\draw[-{Stealth}] (A) --
  node[midway,above] {$f$} (B);
```

The options work together: `midway` selects a position along the path, and `above` selects a side of the path.

```

\begin{tikzpicture}
  \node[draw] (A) at (0,0) {$A$};
  \node[draw] (B) at (5,0) {$B$};

  \draw[-{Stealth}] (A) --
    node[near start,above] {start}
    node[midway,below] {middle}
    node[near end,above] {end}
    (B);
\end{tikzpicture}

```

See



Most mathematical arrows need only one short label. The three-label example is intended to make the positions visible.

3.5.1 Labels on Curved Paths

The label belongs between `to` and the endpoint:

```

\draw[-{Stealth}] (A)
  to[bend left=25]
  node[midway,above] {$f$}
  (B);

```

3.6 Right-Angle Connections

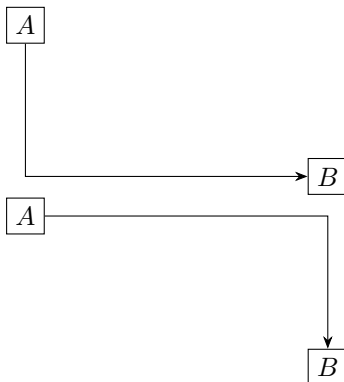
The operators `|-` and `-|` use the same two endpoint names but take different routes.

```
\begin{tikzpicture}
  \node[draw] (A) at (0,2) {$A$};
  \node[draw] (B) at (4,0) {$B$};

  \draw[-{Stealth}] (A) |- (B);
\end{tikzpicture}
```

With `|-`, the path moves horizontally first and vertically second. With `-|`, it moves vertically first and horizontally second.

See



3.7 Worked Project: A Labeled Square

The project combines named nodes, straight arrows, path labels, and one curved comparison arrow.

3.7.1 Stage 1: Place Four Objects

```
\node[draw] (A) at (0,2) {$A$};  
\node[draw] (B) at (4,2) {$B$};  
\node[draw] (C) at (0,0) {$C$};  
\node[draw] (D) at (4,0) {$D$};
```

3.7.2 Stage 2: Add the Square

```
\draw[-{Stealth}] (A) -- node[above] {$f$} (B);  
\draw[-{Stealth}] (A) -- node[left] {$g$} (C);  
\draw[-{Stealth}] (B) -- node[right] {$h$} (D);  
\draw[-{Stealth}] (C) -- node[below] {$k$} (D);
```

3.7.3 Stage 3: Add a Comparison Arrow

```
\draw[-{Stealth}] (A)  
  to[bend left=18]  
  node[midway,above] {$q$}  
  (D);
```

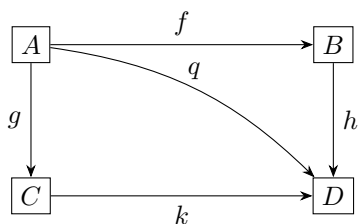

3.7.4 The Complete Picture

```
\begin{tikzpicture}
  \node[draw] (A) at (0,2) {$A$};
  \node[draw] (B) at (4,2) {$B$};
  \node[draw] (C) at (0,0) {$C$};
  \node[draw] (D) at (4,0) {$D$};

  \draw[-{Stealth}] (A) -- node[above] {$f$} (B);
  \draw[-{Stealth}] (A) -- node[left] {$g$} (C);
  \draw[-{Stealth}] (B) -- node[right] {$h$} (D);
  \draw[-{Stealth}] (C) -- node[below] {$k$} (D);

  \draw[-{Stealth}] (A)
    to[bend left=18]
    node[midway,above] {$q$}
    (D);
\end{tikzpicture}
```

See



Design

The four structural arrows remain straight. Only the additional comparison arrow is curved. The difference in routing therefore has a reason.

3.8 Debugging Workshop

3.8.1 Forgetting Parentheses Around a Name

Incorrect:

```
\draw A -- B;
```

Correct:

```
\draw (A) -- (B);
```

3.8.2 Using a Bend with --

Use `to` when a connection needs `bend left` or `bend right`:

```
\draw (A) to[bend left=20] (B);
```

3.8.3 Putting a Label Outside the Path

The path label belongs between the start and endpoint. If it is written after the endpoint, it no longer occupies the intended position on the connection.

3.8.4 Choosing the Wrong Orthogonal Operator

If a right-angle path bends the wrong way, exchange `|-` and `-|`. Do not move the nodes until the routing operator has been checked.

3.9 Exercises

1. Connect three named nodes by two straight lines.
2. Replace the lines by `--{Stealth}` arrows.
3. Draw two opposite curved arrows between the same pair of nodes.
4. Label one arrow near its start, one midway, and one near its end.
5. Place a label above a curved arrow.
6. Draw both `|-` and `-|` between diagonally separated nodes.
7. Create a four-object diagram with labels on all four sides.
8. Add one extra curved arrow without moving the four objects.

3.10 Solutions

3.10.1 Exercises 1–3

```
\node[draw] (A) at (0,0) {$A$};  
\node[draw] (B) at (2.5,0) {$B$};  
\node[draw] (C) at (5,0) {$C$};  
\draw (A) -- (B);  
\draw (B) -- (C);
```

```
\draw[-{Stealth}] (A) -- (B);  
\draw[-{Stealth}] (B) -- (C);
```

```
\draw[-{Stealth}] (A) to[bend left=20] (B);  
\draw[-{Stealth}] (B) to[bend left=20] (A);
```

3.10.2 Exercises 4–6

```
\draw[-{Stealth}] (A) -- node[near start,above] {$f$} (B);
\draw[-{Stealth}] (B) -- node[midway,above] {$g$} (C);
\draw[-{Stealth}] (C) -- node[near end,below] {$h$} (D);
```

```
\draw[-{Stealth}] (A)
  to[bend left=25]
  node[midway,above] {$f$}
  (B);
```

```
\draw[-{Stealth}] (A) |- (B);
\draw[-{Stealth}] (A) -| (B);
```

Exercises 7 and 8 may follow the worked project with different node text and arrow labels.

3.11 ChatGPT Workshop

Ask ChatGPT to preserve node positions unless moving them is part of the task.

1. Add short labels to these TikZ arrows without moving any node.
2. Replace these overlapping straight arrows by the smallest useful `bend left` or `bend right` values.
3. Explain why `|-` and `-|` take different routes between the same endpoints.
4. Check the direction of every arrow and the placement of every path label in this diagram.
5. Convert these raw-coordinate endpoints into connections between named nodes without changing the visible picture.

3.12 Final Checklist

Before continuing to Chapter 4, check that you can

1. connect named nodes with `--`;
2. add a modern **Stealth** arrow tip;
3. use `to` with a restrained bend;
4. place a label at the start, middle, or end of a connection;
5. label a curved path; and
6. distinguish the routes produced by `|-` and `-|`.